

TATE'S THESIS

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ABSTRACT. L -functions are of great interest to number theorists. Key to their study are their meromorphic extensions and functional equations. Hecke defined a class of L -functions analogous to Dirichlet's and used unpleasant methods to give their meromorphic extensions and functional equations [4]. In his thesis, Tate bypassed Hecke's methods by using simple Fourier analysis to derive the same [9]. We give Tate's derivation, but following Kudla [5] (who follows Weil [10],) we reinterpret the pervasive proportionality of distributions. To illustrate the theory, we compute a few concrete examples.

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1. INTRODUCTION

L -functions are of great classical interest, for they contain detailed information about distribution of primes. Obtaining their meromorphic extensions to the complex plane and their functional equations are key tools for studying them. In his thesis, Tate [9] used fairly simple Fourier analysis to obtain these extensions and functional equations for a specific class of L -functions, a significant improvement over the techniques of the time [4]. We will closely follow the exposition in Kudla [5], which follows Weil [10] in reinterpreting the local functional equation.

In Section 2, we will introduce the main players.

In Section 3, we will work out the situation in local contexts. A space of distributions will turn out to be one-dimensional, forcing important distributions to be related by scale-factors. One of these scale factors is, conveniently, the local factor of a classical L -function.

In Section 4, adèles and idèles will be useful global contexts, for we can produce their pictures by multiplying together all local pictures. Therefore, the entire L -function whose factors appeared locally will appear globally. There will be a scale factor which we can compute in terms of this L -function; however, we can also use Poisson summation to compute this scale factor as 1. This gives a functional equation for our L -function.

The reader should note that there only two crucial theoretical points in Sections 3 and 4, namely Lemma 3.4 and Theorem 4.12.

Finally, we will show the relationship to classical pictures in Section 5, computing the functional equations for Dirichlet L -functions and certain Hecke L -functions over $\mathbb{Q}(\sqrt{5})$.

2. MOTIVATION, CHARACTERS

By “valuation” we mean a norm; that is, we write codomains multiplicatively and allow archimedean valuations. By a finite valuation we mean a nonarchimedean valuation, and by an infinite valuation we mean an archimedean valuation.

Notations 2.1. We will use the following notation.

- K is a global field; that is, a finite extension of $\mathbb{Q}$ or $\mathbb{F}_p(t)$.
- If K is a number field, $\mathcal{O}_K$ is its ring of integers.
- K_v is a completion of K with respect to a valuation v .
- $\mathcal{O}_{K_v}$ is the ring of integers in K_v for finite v , $\mathfrak{m}_v$ is its maximal ideal, and π_v is a uniformizer.
- $\mathcal{U}_v$ is the group of elements of K_v^* of norm 1.
- dx is an additive Haar measure on K_v , and d^*x is a multiplicative Haar measure on K_v^* .
- π_v is a uniformizer of K_v for finite v .
- For a finite valuation v , $\mathfrak{p}_v$ is the prime of $\mathcal{O}_K$ inducing it, and for a prime $\mathfrak{p}$ of $\mathcal{O}_K$, $v_{\mathfrak{p}}$ is the valuation induced by it.
- If v is finite, f_v° is the characteristic function of $\mathcal{O}_{K_v}$.

We will tacitly normalize all valuations v so that multiplication by any $\alpha \in K_v$ scales dx by $|\alpha|$; that is, $d(\alpha x) = |\alpha|_v dx$. Thus, d^*x and $\frac{dx}{|x|_v}$ agree up to a constant. If $K_v = \mathbb{C}$, this means $||_v$ is the *square* of the usual Euclidean norm.

2.1. Characters.

Definition 2.2. If G is a locally compact abelian group, a *quasicharacter* of G is a continuous homomorphism $\omega : G \rightarrow \mathbb{C}^*$, and a *character* of G is a quasicharacter whose image lands in S^1 .

Remarks 2.3. If G is compact, any quasicharacter is a character. Quasicharacters and characters form groups under pointwise multiplication.

We write down the following lemma for future reference.

Lemma 2.4. *If G is a compact abelian group, ω is a nontrivial character of G , and dx is a Haar measure on G , $\int_G \omega(x) dx = 0$.*

Proof. Let a be any element of G on which ω is nontrivial. If dx is a Haar measure,

$$\omega(a) \int_G \omega(x) dx = \int_G \omega(ax) dx = \int_G \omega(x) dx,$$

which can only hold if $\int_G \omega(x) dx = 0$. $\square$

Recall that a *Dirichlet character* $\chi \bmod m$ is a character of $(\mathbb{Z}/m)^*$, and χ is *primitive* if it is not the pullback of a Dirichlet character mod n for any n properly dividing m . By pulling back, we can identify primitive Dirichlet characters with characters of $\hat{\mathbb{Z}}^* = \varprojlim (\mathbb{Z}/m\mathbb{Z})^*$. We will further refine this group to the group of idèles and then treat characters on these, though the benefit of this will not become clear until Section 4.

Construction 2.5. For a given character $\omega : K_v \rightarrow S^1$, we can consider all quasicharacters of the form $\omega(x) \cdot |x|_v^s$ for $s \in \mathbb{C}$. Parameterized by s , these form a connected Riemann surface, and the analytic structure we assign this surface is independent of our choice of ω . These partition all quasicharacters.

The picture is this: if v is infinite, these Riemann surfaces are copies of $\mathbb{C}$, and if v is finite, they are copies of the cylinder given by taking $\mathbb{C} \bmod 2\pi i / \log |\pi_v|_v^{-1}$. The characters on this surface are the imaginary axis $\operatorname{Re}(s) = 0$ (a circle if v is finite).

We will now begin to classify characters of K_v^* for an arbitrary completion K_v of K .

Definition 2.6. A quasicharacter ω of K_v^* is *unramified* if it is trivial on $\mathcal{U}_v$; that is, if some complex s exists such that $\omega(x) = |x|_v^s$ for all x . For finite v , a quasicharacter of K_v is unramified if it is trivial on $\mathcal{O}_{K_v}$.

We can classify quasicharacters ω of K_v^* : by compactness of $\mathcal{U}_v$, ω can be written in the form $\omega'(x) \cdot |x|_v^s$ for some *character* ω' and some $s \in \mathbb{C}$. ω' is not well-defined, though its restriction to $\mathcal{U}_v$ is, and s is not well-defined, though its real part is.

Definition 2.7. The *exponent* of ω is $\operatorname{Re}(s)$.

We can now see the following. Our Riemann surfaces are the cosets of the unramified quasicharacters; that is, they correspond to restrictions to $\mathcal{U}_v$. The translates on each Riemann surface of the characters correspond to exponents.

We can classify these even further with the following observation. If v is finite, a character of $\mathcal{U}_v = \varprojlim_n (\mathcal{O}_{K_v}/\mathfrak{m}_v^n)^*$ is the pullback of a character of some $(\mathcal{O}_{K_v}/\mathfrak{m}_v^n)^*$.

Definition 2.8. The *conductor* of a character ω of $\mathcal{U}_v$ is the smallest c such that ω is the pullback of a character of $(\mathcal{O}_{K_v}/\mathfrak{m}_v^c)^*$. If ω is a character of K_v , let the conductor of ω be the conductor of its restriction to $\mathcal{U}_v$.

Thus, we can specify a quasicharacter by specifying a conductor c , a character of $(\mathcal{O}_{K_v}/\mathfrak{m}_v^n)^*$, and the image of a uniformizer π_v ; the first two are finite amounts of data.

The classification in the infinite case is even simpler: if $K_v = \mathbb{R}$, a character of $\mathcal{U}_v$ is of the form $x \mapsto x^n$, where n is a congruence class mod 2, and if $K_v = \mathbb{C}$, a character of $\mathcal{U}_v$ is of the form $x \mapsto x^n$, where $n \in \mathbb{Z}$.

2.2. Restricted direct products. Let $(G_v)_{v \in V}$ be an indexed family of locally compact groups, cofinitely many of which have an open compact subgroup H_v . Their *restricted direct product* $\prod'_v (G_v, H_v)$ is the group of all elements of $\prod_v G_v$ with cofinitely many components in H_v . Topologize this by taking as an open base sets of the form $\prod_v U_v$ intersected with the restricted direct product, where every $U_v \subseteq G_v$ is open and cofinitely many U_v s contain H_v .

We will usually suppress the H_v , writing the product instead as $\prod'_v G_v$.

Now, for any finite collection $v_1, \dots, v_n$ of indices, $\prod_1^n G_{v_i} \times \prod_{v \neq v_1, \dots, v_n} H_v$ is an open subset of $\prod'_v G_v$, and the subspace and product topologies coincide. The subspace topology is locally compact by Tychonoff's theorem, and subsets of this form cover $\prod'_v G_v$. Therefore $\prod'_v G_v$ is again locally compact.

Every G_v embeds into $\prod'_v G_v$ by sending g_v to the tuple whose v th coordinate is g_v and whose other coordinates are 1; call this embedding ι_v .

For $v \in V$ let ω_v be a (quasi)character of G_v , and assume cofinitely many ω_v s are unramified. We can now define a (quasi)character of $\prod'_v G_v$ by the rule $(x_v) \mapsto \prod_v \omega_v(x_v)$.

Proposition 2.9. *This gives a bijection between (quasi)characters of $\prod'_v G_v$ and such collections (ω_v) of (quasi)characters.*

The inverse map is obvious: define our ω_v s by $\omega_v(x_v) = \omega(x_v)$, where we identify $x_v \in G_v$ with its image under ι_v . All that remains to be checked is that these are indeed inverse bijections; for a proof, see Lemmas 3.2.1 and 3.2.2 of [9].

Definition 2.10. If ω is a quasicharacter of $\prod'_v G_v$, let ω_v be its factor at v , and call (ω_v) the *local factors* of ω .

2.3. Adèles and idèles.

Examples 2.11. We will only consider two examples of restricted direct products.

- The restricted direct product $\prod'_v (K_v, \mathcal{O}_{K_v})$ of the completions of K is called the *ring of adèles* of K and denoted $\mathbb{A}_K$. As the name suggests, this is a ring, with product inherited from the unrestricted product $\prod_v K_v$. (Its topology is *not* inherited from the unrestricted product.) Further, we can regard K as a discrete subring of $\mathbb{A}_K$ with the diagonal embedding $x \mapsto (x)_v$.
- The restricted direct product $\prod'_v (K_v^*, \mathcal{U}_v)$ is the *group of idèles* of K and denoted $\mathbb{A}_K^*$. Just like in the adèlic case, we can regard K^* as a discrete subgroup.

This is the group of units in $\mathbb{A}_K$, but with a finer topology than the subspace topology. It is, however, the coarsest topology refining the subspace

topology under which inversion is continuous; equivalently, the topology inherited from the embedding

$$\alpha \mapsto (\alpha, \alpha^{-1}) : \mathbb{A}_K^* \rightarrow \mathbb{A}_K \times \mathbb{A}_K.$$

This construction has two points. First, $\mathbb{A}_K$ and $\mathbb{A}_K^*$ are locally compact, so one can do harmonic analysis on them. Second, K sits discretely in $\mathbb{A}_K$ and K^* sits discretely in $\mathbb{A}_K^*$; see [2, § 14] and [2, § 16]. However, the following results show that this is a precarious situation.

Theorem 2.12 (Weak approximation theorem). *If V is any finite set of valuations of K , K is dense in $\prod_{v \in V} K_v$.*

Theorem 2.13 (Strong approximation theorem). *If we omit any one of K 's completions, K is dense in the restricted direct product $\prod'_v K_v$.*

See [2, § 6] and [2, § 15] for proofs.

Let K be a number field. If (x_v) is an idèle, for every finite v let e_v be the unique integer satisfying $x_v/\pi_v^{e_v} \in \mathcal{O}_v$. Cofinitely many e_v s are 0, so $\prod_{v \text{ nonarch.}} \mathfrak{p}_v^{e_v}$ is a fractional ideal of $\mathcal{O}_K$.

Definition 2.14. $\prod_{v \text{ nonarch.}} \mathfrak{p}_v^{e_v}$ is the fractional ideal *generated by* (x_v) .

$(x_v) \mapsto \prod_{v \text{ nonarch.}} \mathfrak{p}_v^{e_v}$ is an epimorphism.

Define a projection π_v from $\mathbb{A}_{\mathbb{Q}}^*$ to $\widehat{\mathbb{Z}}^*$ as follows. If $(x_v) \in \mathbb{A}_{\mathbb{Q}}^*$, cofinitely many x_v s are in their respective $\mathbb{Z}_{\mathfrak{p}_v}$. Therefore a unique rational number α exists such that $\alpha(x_v) = (\alpha x_v)$ has all finite components in their respective $\mathbb{Z}_{\mathfrak{p}_v}$ and αx_{∞} is nonnegative, namely $\text{sgn}(x_{\infty}) \cdot \prod_{v \neq \infty} |x_v|_v^{-1}$. Now $(\alpha \cdot x_v)_{v \neq \infty}$ is an element of $\prod_{v \neq \infty} \mathbb{Z}_{\mathfrak{p}_v}^* = \widehat{\mathbb{Z}}^*$. Let this be $\pi_v((x_v))$.

This is a homomorphism whose kernel contains $\mathbb{Q}^*$.

Construction 2.15. If $\chi : (\mathbb{Z}/m)^* \rightarrow S^1$ is a Dirichlet character, we can pull back χ^{-1} under the projection $\mathbb{A}_{\mathbb{Q}}^* \rightarrow \widehat{\mathbb{Z}}^* \rightarrow (\mathbb{Z}/m)^*$ and obtain an idèlic character ω trivial on $\mathbb{Q}^*$.

The reason for using χ^{-1} is as follows. If p is a prime, consider the idèle $\iota_{v_p}(p)$ whose p th component is p and whose other components are 1. If p is coprime to the conductor of χ , $\omega_p(p) = \chi(\tilde{p})$ holds, where $\tilde{p}$ is the residue of p mod the conductor of χ .

For a Dirichlet character χ , define its *Dirichlet L -function*

$$(2.16) \quad L(s, \chi) = \sum_{n \geq 1} \frac{\chi(n)}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - \chi(p) \cdot p^{-s}}$$

for $\text{Re}(s) > 1$. This will turn out to be (up to a meromorphic factor) a constant of proportionality between two adèlic distributions, and the factors of the Euler product will turn out to be constants of proportionality between p -adic distributions. We will also extend $L(s, \chi)$ meromorphically to the entire plane and use adèlic Fourier analysis to derive a functional equation.

The same theory will apply similarly for Hecke characters and zeta-functions, but to avoid overcomplicating things we will explain this after completing our discussion of Dirichlet L -functions.

2.4. Abelian Fourier analysis. Here we will give, but not prove, the main results of abelian Fourier analysis. The reader wishing to know more may consult [8, § 3].

Let G be a locally compact abelian group with a Haar measure dx .

Definition 2.17. The *Pontryagin dual* $\hat{G}$ of G is defined as the group of characters of G , with the compact-open topology.

Theorem 2.18 (Pontryagin duality). *There is a natural isomorphism $G \cong \hat{\hat{G}}$.*

That is, $\text{Hom}_{\text{cts}}(-, S^1)$ is an involution of the category of locally compact abelian groups.

Pontryagin duality forms the correct context for abstract abelian Fourier analysis. In particular, we will need the following fact.

Definition 2.19. If $f \in L^2(G) \cap L^1(G)$, define the *Fourier transform* $\hat{f} : \hat{G} \rightarrow \mathbb{C}$ of f by $\hat{f}(\xi) = \int_G f(y)\xi(y) dy$.

Theorem 2.20. *The Fourier transform extends by continuity to give an isomorphism of $L^2(G)$ with $L^2(\hat{G})$. There is a unique Haar measure on $\hat{G}$ so that the Fourier transform is unitary and $\hat{\hat{f}}(x) = f(-x)$ holds; this measure is said to be dual to dx .*

3. LOCAL THEORY

3.1. Setting the stage. We will define the topological vector space $S(K_v)$ of *Schwartz-Bruhat functions* on K_v as follows. These constructions are largely unimportant; we just need a well-behaved space of test functions. For example, [9] uses a suitable subspace of L^1 . That being said, $S(K_v)$ is particularly convenient.

If v is finite, define $S(K_v)$ to be the $\mathbb{C}$ -vector space of compactly-supported locally-constant functions $K_v \rightarrow \mathbb{C}$, frivolously topologized as the direct limit of its finite-dimensional subspaces. Thus any linear map out of $S(K_v)$ is automatically continuous.

If $K_v = \mathbb{R}$, for $f \in C^\infty(\mathbb{R})$ and $N, i \in \mathbb{N}$ define

$$\|f\|_{(N,i)} = \sup_x \left((1 + |x|)^N \cdot \left| f^{(i)}(x) \right| \right).$$

Let $S(\mathbb{R})$ be the space of f such that every $\|f\|_{(N,i)}$ is finite; that is, f and all its derivatives are dominated by any power of x . The $\|\cdot\|_{(N,i)}$ s are now seminorms on $S(\mathbb{R})$; let these generate our topology on $S(\mathbb{R})$.

If $K_v = \mathbb{C}$, define $S(\mathbb{C})$ as in the case $K = \mathbb{R}$, but with the seminorms given by

$$\|f\|_{(N,i,j)} = \sup_x \left((1 + |x|)^N \cdot \left| \partial_x^i \partial_{\bar{x}}^j f(x) \right| \right).$$

Both of these are Fréchet spaces; see [3, Prop 8.2].

Definition 3.1. A *tempered distribution* on K_v is a continuous linear functional on $S(K_v)$; that is, a continuous linear map $S(K_v) \rightarrow \mathbb{C}$.

Examples 3.2. The point mass δ_0 defined by $\langle \delta_0, f \rangle = f(0)$ is a distribution. More generally, if dx' is a Radon measure on K_v and v is finite, integration against dx' is a distribution. This is also true for infinite v if there is some compact K_0 and real $a, c > 0$ such that, on compact sets containing K_0 , dx' is dominated by $c|x|_v^a dx$.

Denote the space of tempered distributions by $S(K_v)'$. For $\lambda \in S(K_v)'$ and $f \in S(K_v)$ we will write $\lambda(f)$ as $\langle \lambda, f \rangle$.

We can define an action of K_v^* on $S(K_v)'$ with $\langle a\lambda, f \rangle = \langle \lambda, f(xa^{-1}) \rangle$. Given a quasicharacter ω of K_v^* , we can define another action of K_v^* on $S(K_v)'$ by multiplying by $\omega(a)$.

Definition 3.3. If λ is a distribution on which these two actions coincide, we say λ is an ω -eigendistribution, and denote the space of ω -eigendistributions by $S'(\omega)$.

The uncomfortable reader may be convinced this terminology is sound as follows. If λ is an eigendistribution of every $\lambda \mapsto a\lambda$, as defined by the first action, trivially the eigenvalues $\omega(a)$ form a quasicharacter, therefore λ is an ω -eigendistribution.

The crux of the local theory is the following.

Lemma 3.4. *For any quasicharacter ω , $S'(\omega)$ is one-dimensional.*

On our quest to prove this, we will meet the factors of the Euler products of classical L -function as scale factors. Passing to the global (idèlic) picture will multiply all of these together, yielding the entire L -function as a scale factor.

3.2. Lemma 3.4's proof. Lemma 3.4 is proven as (coincidentally) Theorem 3.4 of [5]. The arguments we will use here are essentially the same, though we will use less fancy language at the cost of our setup being less canonical. Our topologies aren't well-behaved enough to make such grandiose claims as "a distribution decomposes uniquely as a distribution supported at 0 plus a distribution supported away from 0," but we will set up a slightly more careful version of this claim.

If v is finite, let f be any Schwartz-Bruhat function with $f(0) \neq 0$. There is a short exact sequence

$$0 \longrightarrow C_c^\infty(K_v^*) \longrightarrow S(K_v) \xrightarrow{\text{ev}_0} \mathbb{C} \longrightarrow 0,$$

where $C_c^\infty(K_v^*)$ is the space of functions in $S(K_v)$ supported away from 0. We can split this, writing $S(K_v) = C_c^\infty(K_v^*) \oplus \mathbb{C}f$. A distribution is thus defined by its restriction to $C_c^\infty(K_v^*)$ and its value on f .

Define ω -eigendistributions on K_v^* in the obvious way, namely as linear $\lambda : C_c^\infty(K_v^*) \rightarrow \mathbb{C}$ for which $\langle \lambda, f(xa^{-1}) \rangle = \langle \omega(a)\lambda, f \rangle$ for all $a \in K_v^*$. The following facts are clear.

- ω -eigendistributions on K_v^* are spanned by $\omega(x)d^*x$.
- Any ω -eigendistribution in $S'(\omega)$ restricts to an ω -eigendistribution on K_v^* .
- δ_0 spans all distributions supported at 0, and is an ω -eigendistribution if and only if ω is trivial.

We can now verify Lemma 3.4 by assuming λ is an ω -eigendistribution and asking what this forces about $\langle \lambda, f \rangle$. This will essentially reduce to a formal game.

Case 1: v is finite and ω is ramified.

Let λ be an ω -eigendistribution. Then some $c \in \mathbb{C}$ must exist such that λ coincides with $c\omega(x)d^*x$ on $C_c^\infty(K_v^*)$. We will check that this determines λ on all remaining functions.

Let a be any element of $\mathcal{U}_v$ that ω is nontrivial on. That λ is an ω -eigendistribution forces

$$\langle \lambda, f_v^o \rangle = \omega(a) \langle \lambda, f_v^o \rangle$$

to hold, so $\langle \lambda, f_v^o \rangle = 0$. It follows that $\langle \lambda, f_v^o(x\pi_v^{-n}) \rangle = 0$ for all $n \in \mathbb{Z}$; that is, λ is 0 on the characteristic function of any $\mathfrak{m}_v^n$.

However, as any $f \in S(K_v)$ is constant on a neighborhood $\mathfrak{m}_v^n$ of 0, f differs by one of these characteristic functions from a function supported away from 0; this implies

$$\langle \lambda, f \rangle = \langle \lambda, f - f(0)f_v^o(x\pi_v^{-n}) \rangle = c \int_{K_v^* - \mathfrak{m}_v^n} f(x)\omega(x) d^*x.$$

This shows that $S'(\omega)$ is at most one-dimensional, for λ is determined by c . Further, we can now check $S'(\omega)$ is populated by defining a nonzero ω -eigendistribution

$$\langle z_0(\omega), f \rangle := \int_{K_v^* - \mathfrak{m}_v^n} f(x)\omega(x) d^*x.$$

This is independent of our choice of n , so long as it's large enough for f to be constant on $\mathfrak{m}_v^n$. We can reinterpret $z_0(\omega)$ in a more natural way as the principal value integral

$$\langle z_0(\omega), f \rangle = \text{PV} \int_{K_v^*} f(x)\omega(x) d^*x = \lim_{n \rightarrow \infty} \int_{K_v^* - \mathfrak{m}_v^n} f(x)\omega(x) d^*x.$$

Case 2: v is finite and ω is unramified and nontrivial.

We proceed similarly. Let λ be an ω -eigendistribution. Again, some $c \in \mathbb{C}$ exists such that λ coincides with $c\omega(x) d^*x$ on $C_c^\infty(K_v^*)$. As f is Schwartz-Bruhat, $f(x) - f(x\pi_v^{-1})$ is supported away from 0, implying

$$\langle \lambda, f(x) - f(x\pi_v^{-1}) \rangle = c \int (f(x) - f(x\pi_v^{-1})) \omega(x) d^*x.$$

As λ is an ω -eigendistribution, we also have

$$\langle \lambda, f(x) - f(x\pi_v^{-1}) \rangle = \langle \lambda, f \rangle - \langle \lambda, f(x\pi_v^{-1}) \rangle = (1 - \omega(\pi_v)) \langle \lambda, f \rangle.$$

This gives us

$$\langle \lambda, f \rangle = c(1 - \omega(\pi_v))^{-1} \int (f(x) - f(x\pi_v^{-1})) \omega(x) d^*x,$$

showing that $S'(\omega)$ is at most one-dimensional. By defining

$$\langle z_0(\omega), f \rangle = \int (f(x) - f(x\pi_v^{-1})) \omega(x) d^*x,$$

we obtain a nonzero ω -eigendistribution.

Case 3: ω is trivial.

Let λ be an ω -eigendistribution, i.e. let λ be K_v^* -invariant, and let c be as usual. We have

$$\langle \lambda, f_v^o \rangle = \langle \lambda, f_v^o(x\pi_v^{-1}) \rangle,$$

and

$$0 = \langle \lambda, f_v^o(x) - f_v^o(x\pi_v^{-1}) \rangle = c \int_{\mathcal{U}_v} d^*x$$

follows. This forces c to be 0; that is, λ is trivial on functions supported away from 0. Therefore λ is a scalar multiple of δ_0 . Conversely, any scalar multiple of δ_0 is an ω -eigendistribution.

Notice that, if we define $z_0(1)$ as in Case 2, $z_0(1) = \delta_0 \int_{\mathcal{U}_v} d^*x$.

Case 4: v is infinite.

We summarize, but do not prove, the results in this case. If we define

$$L_v(\omega) = \begin{cases} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) & \text{if } K_v = \mathbb{R} \text{ and } \omega(x) = |x|^s \\ \pi^{-\frac{s+1}{2}} \Gamma\left(\frac{s+1}{2}\right) & \text{if } K_v = \mathbb{R} \text{ and } \omega(x) = \text{sgn}(x) \cdot |x|^s \\ (2\pi)^{1-s+\frac{|n|}{2}} \Gamma\left(s + \frac{|n|}{2}\right) & \text{if } K_v = \mathbb{C} \text{ and } \omega(x) = (x/|x|^{1/2})^n |x|^s \end{cases},$$

then we can holomorphically choose a generator $z_0(\omega)$ for $S'(\omega)$ whose restriction to K_v^* is $L_v(\omega)^{-1} \omega(x) d^*x$. We refer the reader to [1, Prop 3.1.8], [5, Pages 121-122], and [10] for more details.

3.3. Recap. For all ω , we have just defined a distribution $z_0(\omega)$, which gives a nonzero generator for the one-dimensional vector space $S'(\omega)$. Another key point is that we have parameterized $z_0(\omega)$ *holomorphically* in ω . This allows us to conclude that certain constants of proportionality are holomorphic.

Unless ω is the trivial character in the finite case, x^n in the real case, or $(x/|x|^{1/2})^{n_1} x^{n_2}$ in the complex case, the restriction of $z_0(\omega)$ to K_v^* is nonzero. This means there is another generator, call it $z(\omega)$, whose restriction to K_v^* is $\omega(x) d^*x$. By our calculations in proving Lemma 3.4, we obtain

$$(3.5) \quad z(\omega) = z_0(\omega) \cdot L_v(\omega),$$

where

$$(3.6) \quad L_v(\omega) = \begin{cases} \frac{1}{1-\omega(\pi_v)} & \text{if } v \text{ is finite and } \omega \text{ is unramified} \\ 1 & \text{if } v \text{ is finite and } v \text{ is ramified} \\ \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) & \text{if } K_v = \mathbb{R} \text{ and } \omega(x) = |x|^s \\ \pi^{-\frac{s+1}{2}} \Gamma\left(\frac{s+1}{2}\right) & \text{if } K_v = \mathbb{R} \text{ and } \omega(x) = \text{sgn}(x) \cdot |x|^s \\ (2\pi)^{1-s+\frac{|n|}{2}} \Gamma\left(s + \frac{|n|}{2}\right) & \text{if } K_v = \mathbb{C} \text{ and } \omega(x) = (x/|x|^{1/2})^n |x|^s \end{cases}.$$

Remark 3.7. Notice $z(\omega)$ is integration against $\omega(x) d^*x$ if ω has positive exponent; that is, if this integral converges. Therefore $z(\omega)$ can be viewed as a meromorphic extension of $\omega(x) d^*x$. $L_v(\omega)^{-1}$ is the factor necessary to eliminate the zeroes and poles of $z(\omega)$, yielding a nonzero holomorphic family of distributions $z_0(\omega)$. That $z_0(\omega)$ be scaled *exactly* as we have defined it is also critical; see Remark 4.1.

Remark 3.8. If χ is a Dirichlet character, we have already described (Construction 2.15) how to pull back χ^{-1} to obtain a character of $\mathbb{A}_{\mathbb{Q}}^*$. If ω is this character's factor at a prime p , note $L_{v_p}(\omega \cdot ||_{v_p}^s) = \frac{1}{1-\chi(p) \cdot p^{-s}}$ is the factor at p of the Euler product $L(s, \chi)$. In Section 5, this will be the connection to Dirichlet (and Hecke) L -functions.

3.4. Local Fourier analysis and the local functional equation. Much like in the real case, if K_v is a local field and $\psi : K_v \rightarrow S^1$ is a nontrivial character, $y \mapsto \psi(y \cdot -)$ gives an isomorphism of K_v with its character group; that is, K_v is self-dual. Identifying K_v with its dual in this way, the Fourier transform of a function $f \in L^1(K_v) \cap L^2(K_v)$ is given by $\hat{f}(\xi) = \int_{K_v} f(x)\psi(x\xi) dx$. The Fourier transform is not only an automorphism of $L^2(K_v)$, but also of $S(K_v)$. The choice of dx such that $\hat{\hat{f}} = f(-x)$ holds is called *self-dual* with respect to ψ . See [8, Sec 3.3] and [8, Prop 7-1] for more details.

For the rest of this subsection, fix a character $\psi : K_v \rightarrow S^1$, and define the Fourier transform using the corresponding self-dual measure dx .

Definitions 3.9. If λ is a distribution, its *Fourier transform* $\hat{\lambda}$ is defined by $\langle \hat{\lambda}, f \rangle = \langle \lambda, \hat{f} \rangle$. If ω is a character of K_v^* , its *shifted dual* is $\hat{\omega} = ||_v \cdot \omega^{-1}$.

If λ is an ω -eigendistribution, $\hat{\lambda}$ is a $\hat{\omega}$ -eigendistribution, for if $a \in K_v^*$,

$$\begin{aligned} \langle \hat{\lambda}, f(xa^{-1}) \rangle &= \langle \lambda, \widehat{f(xa^{-1})} \rangle = \langle \lambda, |a|_v \hat{f}(xa) \rangle \\ &= |a|_v \langle \lambda, \hat{f}(xa) \rangle = |a|_v \cdot \omega(a^{-1}) \langle \lambda, \hat{f} \rangle = |a|_v \cdot \omega^{-1}(a) \langle \hat{\lambda}, f \rangle. \end{aligned}$$

Combining this with Lemma 3.4 and the fact that z_0 is holomorphic in ω , we obtain the following.

Theorem 3.10 (Local functional equation). *Some holomorphic nonzero factor $\varepsilon_v(\omega, \psi)$ exists such that $\varepsilon_v(\omega, \psi) \cdot z_0(\omega) = \widehat{z_0(\hat{\omega})}$, and if ω is nontrivial some meromorphic $\gamma_v(\omega, \psi)$ exists such that $\gamma_v(\omega, \psi) \cdot z(\omega) = \widehat{z(\hat{\omega})}$. These are related by the equation $\gamma_v(\omega, \psi) = \varepsilon_v(\omega, \psi) \cdot \frac{L_v(\hat{\omega})}{L_v(\omega)}$.¹*

Proposition 3.11. *For $y \in K_v$, $\varepsilon_v(\omega, \psi(-\cdot y)) = |y|_v^{1/2} \omega(y) \varepsilon_v(\omega, \psi)$.*

Proof. Work through definitions. $\square$

Therefore it suffices to compute $\varepsilon_v(\omega, \psi)$ for a single ψ .

To compute $\varepsilon_v(\omega, \psi)$, we need only choose a test function at which to evaluate $z_0(\omega)$ and $z_0(\hat{\omega})$. First, we will make some necessary calculations about dy and f_v^o .

If v is finite, define the conductor ν of ψ by the condition that $\mathfrak{m}_v^\nu$ is the kernel of ψ . By definition,

$$\widehat{f_v^o}(x) = \int_{\mathcal{O}_{K_v}} \psi(xy) dy.$$

By Lemma 2.4, this works out to be $\widehat{f_v^o}(x) = f_v^o(x\pi_v^{-\nu}) \int_{\mathcal{O}_{K_v}} dy$. Similarly, $\widehat{\widehat{f_v^o}}(x) = f_v^o \cdot |\pi_v|_v^\nu \left(\int_{\mathcal{O}_{K_v}} dy \right)^2$; for Fourier inversion to hold, dy must assign $\mathcal{O}_{K_v}$ the measure $|\pi_v|_v^{-\nu/2}$, so $\widehat{f_v^o}(x) = |\pi_v|_v^{-\nu/2} f_v^o(x\pi_v^{-\nu})$.

Now we actually compute $\varepsilon_v(\omega, \psi)$. If v is finite and ω is unramified, we see that

¹Tate proved this differently, using the formula $\langle z(\omega), f \rangle \langle z(\hat{\omega}), \hat{g} \rangle = \langle z(\hat{\omega}), \hat{f} \rangle \langle z(\omega), g \rangle$ to establish this functional equation on the strip $0 < \text{Re}(s) < 1$ and using it to extend z . See [9, Thm 2.4.1].

$$\begin{aligned}
(3.12) \quad \langle z_0(\omega), f_v^o \rangle &= \int_{K_v^*} (f_v^o(x) - f_v^o(x\pi_v^{-1})) \omega(x) d^*x \\
&= \int_{\mathcal{U}_v} d^*x + \int_{K_v^* - \mathcal{U}_v} 0 d^*x = \int_{\mathcal{U}_v} d^*x.
\end{aligned}$$

As $z_0(\hat{\omega})$ is an $\hat{\omega}$ -eigendistribution,

$$\langle \widehat{z_0(\hat{\omega})}, f_v^o \rangle = \langle z_0(\hat{\omega}), |\pi_v|_v^{\nu/2} f_v^o(x\pi_v^{-\nu}) \rangle = \hat{\omega}(\pi_v^\nu) |\pi_v|_v^{\nu/2} \langle z_0(\hat{\omega}), f_v^o \rangle,$$

and $\langle z_0(\omega), f_v^o \rangle = \langle z_0(\hat{\omega}), f_v^o \rangle$ directly from the definition of z_0 ; therefore $\varepsilon_v(\omega, \psi) = \omega(\pi_v^{-\nu}) \cdot |\pi_v|_v^{-\nu/2}$.

The other cases are similar; we briefly describe them. If v is finite and ω is ramified, let c be the conductor of ω . We use the test function

$$g_\omega(x) = \begin{cases} \psi(x) & \text{if } x \in \mathfrak{m}_v^{\nu-c} \\ 0 & \text{else} \end{cases},$$

whose Fourier transform is given by

$$\hat{g}_\omega(\xi) = \begin{cases} |\pi_v|_v^{\nu/2-c} & \text{if } \xi \equiv 1 \pmod{\mathfrak{m}_v^c} \\ 0 & \text{else} \end{cases}.$$

For the case $K_v = \mathbb{R}$, we will only give $\varepsilon_v(\omega, \psi)$ for the character $\psi(x) = e^{-2\pi i \cdot x}$. If $\omega(-1) = 1$, use the test function $e^{-\pi \cdot x^2}$, and if $\omega(-1) = -1$, use the test function $x \cdot e^{-\pi \cdot x^2}$.

For the case $K_v = \mathbb{C}$, we will only give $\varepsilon_v(\omega, \psi)$ for the character $\psi(x) = e^{-4\pi i \cdot \text{Re}(x)}$ and let $n \in \mathbb{Z}$ be the integer such that $\omega(x) = x^n$ on S^1 . If $n \geq 0$, use the test function $\bar{x}^{|n|} e^{-2\pi \cdot |x|}$, and if $n \leq 0$ use $x^{|n|} e^{-2\pi \cdot |x|}$.

For more details on the computation of ε_v , see [5, Prop 3.8] or [9, Sec 2.5]. (Note that Tate's factor ρ is the *inverse* of our factor γ .) We now list the results:

$$(3.13) \quad \varepsilon_v(\omega, \psi) = \begin{cases} \omega(\pi_v^{-\nu}) \cdot |\pi_v|_v^{-\nu/2} & \text{if } v \text{ is finite and } \omega \text{ is unramified} \\ \int_{\pi_v^{\nu-c} \mathcal{U}_v} \omega^{-1}(x) \psi(x) dx & \text{if } v \text{ is finite and } \omega \text{ is ramified} \\ & \text{with conductor } c \\ 1 & \text{if } K_v = \mathbb{R}, \omega(-1) = 1, \\ & \text{and } \psi(x) = e^{-2\pi i \cdot x} \\ -i & \text{if } K_v = \mathbb{R}, \omega(-1) = -1, \\ & \text{and } \psi(x) = e^{-2\pi i \cdot x} \\ (-i)^{|n|} & \text{if } K_v = \mathbb{C}, \omega(x) = x^n \text{ on } S^1, \\ & \text{and } \psi(x) = e^{-4\pi i \cdot \text{Re}(x)} \end{cases}.$$

The remaining cases follow from Proposition 3.11. We can rewrite the case where v is finite and ω is unramified more conveniently as the Gauss sum

$$(3.14) \quad \varepsilon_v(\omega, \psi) = \omega(\pi_v^{c-\nu}) |\pi_v|_v^{\nu/2} \sum_{x \in (\mathcal{O}_{K_v}/\mathfrak{m}_v^c)^*} \omega^{-1}(x) \psi(\pi_v^{\nu-c} x).$$

4. GLOBAL THEORY

4.1. From local analysis to global analysis. We have already seen how local characters multiply to give global characters; here we will do the same for Schwartz-Bruhat functions, distributions, and measures. We will point out that most of this is to work out a framework in which we can multiply the local instances of (3.5) and Theorem 3.10. For more details and proofs, we refer the reader to [9, § 3] and [5, Sec 4].

Define a topological vector space $S(\mathbb{A}_K)$ of Schwartz-Bruhat functions on $\mathbb{A}_K$ as the “restricted tensor product” of all of our $S(K_v)$ s; that is, as the space spanned by all formal products $\bigotimes_v f_v$, where every f_v is in $S(K_v)$, cofinitely many f_v s are f_v^o , and linearity in every f_v is imposed. Topologize this as the direct limit of the tensor products of finitely many $S(K_v)$ s. Notice every element $\sum_i \bigotimes_v f_{i,v}$ of $S(\mathbb{A}_K)$ defines a real function $(x_v) \mapsto \sum_i \prod_v f_{i,v}(x_v)$ on $\mathbb{A}_K$. This assignment is easily seen to be injective, so we will identify $S(\mathbb{A}_K)$ with a space of functions of $\mathbb{A}_K$.

As before, define a tempered distribution as a continuous linear functional on $S(\mathbb{A}_K)$. A tempered distribution is of the form $\bigotimes_v f_v \mapsto \prod_v \langle \lambda_v, f_v \rangle$, where every λ_v is in $S(K_v)'$ and cofinitely many λ_v s are trivial on f_v^o ; see [5, Lem 4.1]. Denote this distribution $\prod_v \lambda_v$.

To multiply Haar measures, let $\prod'_v G_v$ be a restricted direct product, and choose Haar measures dx_v on every G_v of K such that cofinitely many $\mathcal{O}_{K_v}$ s are assigned measure 1. Define a measure “ $dx = \prod_v dx_v$ ” on $\prod'_v G_v$ as follows. Say we have Borel $X_v \subseteq K_v$, cofinitely many of which are $\mathcal{O}_{K_v}$. Give $\prod_v X_v$ the measure $\prod_v \int_{X_v} dx_v$. Extend by declaring dx to be a Radon measure; dx is now a Haar measure. For a more rigorous construction, see [9, Sec 3.3].

Remark 4.1. By (3.12), the normalizations required to multiply local Haar measures on K_v^* and to multiply our local z_0 s coincide: we need cofinitely many $\mathcal{U}_v$ s to be assigned measure 1. This is not a coincidence. For the surface of unramified ω , the holomorphic family $z_0(\omega)$ of distributions is determined up to a nonzero holomorphic factor by the requirement that it be a basis element of $S'(\omega)$. This holomorphic factor is fixed by the requirement that $\langle z_0(\omega), f_v^o \rangle = 1$ for all unramified ω . That is, we deliberately scaled $z_0(\omega)$ to make these conditions coincide. From now on, we will fix measures d^*x_v assigning cofinitely many $\mathcal{U}_v$ s measure 1.

Notation 4.2. Let dx be a Haar measure on $\mathbb{A}_K$, and let d^*x be a Haar measure on $\mathbb{A}_K^*$ constructed as a product of the local measures we just fixed.

Integrating Schwartz-Bruhat functions on $\mathbb{A}_K$ is easy: if $\bigotimes_v f_v$ is a generator of $S(\mathbb{A}_K)$,

$$(4.3) \quad \int_{\mathbb{A}_K} \bigotimes_v f_v dx = \prod_v \int_{K_v} f_v dx_v.$$

We have a global “absolute value” that serves the same purpose as our local ones. Define $|| : \mathbb{A}_K^* \rightarrow \mathbb{R}^*$ by $|(x_v)| = \prod_v |x_v|_v$; that is, let $||$ be the quasicharacter whose local factors are $||_v$. As a corollary of our construction of a global Haar measure from local ones, we see that $d(ax) = |a|dx$ for all idèles a , therefore d^*x and $\frac{dx}{|x|}$ agree up to a constant.

Notations 4.4. We collect all our global notation in one place so we don't lose the reader.

- $\mathbb{A}_K$ is the restricted direct product $\prod'_v K_v$.
- $\mathbb{A}_K^*$ is the restricted direct product $\prod'_v K_v^*$.
- dx is a Haar measure on $\mathbb{A}_K$.
- d^*x is a Haar measure on $\mathbb{A}_K^*$.
- $||$ is the global absolute value we just defined.

We will also need the following.

- $\mathcal{U}_{\mathbb{A}_K}$ is group of idèles of absolute value 1.

4.2. Remarks about characters of $\mathbb{A}_K^*$.

Proposition 4.5. $||$ splits after restricting to its image.

Proof. If K is a number field, K has an infinite valuation v , from which one can construct a splitting of $||$: send $x \in (0, \infty)$ to $\iota_v(x)$ if $K_v = \mathbb{R}$ or $\iota_v(\sqrt{x})$ if $K_v = \mathbb{C}$. If K has positive characteristic, the image of $||$ is isomorphic to $\mathbb{Z}$, so the conclusion follows by abstract nonsense. $\square$

Rephrasing our characterization of local quasicharacters slightly, pick a splitting $\mathbb{A}_K^* \cong \mathcal{U}_{\mathbb{A}_K} \oplus \text{im } ||$, and if $x \in \mathbb{A}_K^*$, let $\tilde{x} \in \mathcal{U}_{\mathbb{A}_K}$ be the first component of x in this splitting. For a quasicharacter ω of $\mathbb{A}_K^*$, we might hope to write $\omega = \omega'(\tilde{x}) \cdot |x|^s$ for some character ω' of $\mathcal{U}_{\mathbb{A}_K}$ and let the exponent of ω be $\text{Re}(s)$. Unfortunately, $\mathcal{U}_{\mathbb{A}_K}$ is not compact, so we cannot always do this. However, $\mathcal{U}_{\mathbb{A}_K}/K^*$ is compact. One sees this by constructing a fundamental domain for K^* in $\mathcal{U}_{\mathbb{A}_K}$ whose closure is compact by Tychonoff's theorem; this closure now surjects to $\mathcal{U}_{\mathbb{A}_K}/K^*$ under the quotient map, forcing $\mathcal{U}_{\mathbb{A}_K}/K^*$ to be compact. The construction of this fundamental domain, while beautiful, would lead us too far astray; the reader may consult [9, Sec 4.3].

Therefore, if ω is a character which is trivial on K^* , the restriction of ω to $\mathcal{U}_{\mathbb{A}_K}$ is in fact a character, so we can write $\omega = \omega'(\tilde{x}) \cdot |x|^s$ and define the exponent of ω to be $\text{Re}(s)$. This is independent of our decomposition $\mathbb{A}_K^* \cong \mathcal{U}_{\mathbb{A}_K} \oplus \text{im } ||$.

We will restrict our attention to these characters. (The actual reason for doing so is to make Theorem 4.12 hold).

Convention 4.6. A character of $\mathbb{A}_K^*$ is assumed to be trivial on K^* .

As an important note, $||$ is still a quasicharacter; that is, $|x| = 1$ for $x \in K^*$. We can show this by using the norm to reduce this to the cases $K = \mathbb{Q}$ and $K = \mathbb{F}_p(t)$, which we can directly check, or can show this by constructing a fundamental domain D for K in $\mathbb{A}_K$ and showing that D and xD have the same measure.

We can now construct a picture of global quasicharacters much like the one for local quasicharacters. Given a global character ω , the space of all quasicharacters of the form $\omega(x)|x|^s$ for $s \in \mathbb{C}$ forms a Riemann surface under this parameterization; this is a cylinder if K has positive characteristic and a plane if K is a number field. This Riemann surface structure is independent of our choice of representative character ω ; the characters correspond to the imaginary axis $\text{Re}(s) = 0$. The translates of this imaginary axis (a circle if K has positive characteristic) correspond to the different exponents of quasicharacters.

4.3. From local eigendistributions to global eigendistributions. For a character ω of $\mathbb{A}_K^*$, define two actions of $\mathbb{A}_K^*$ on the space $S(\mathbb{A}_K)'$ of distributions by defining $a\lambda$ as $\omega(a)\lambda$ or by $\langle a\lambda, f \rangle = \langle \lambda, f(xa^{-1}) \rangle$, and define an ω -eigendistribution

as a distribution on which these two actions coincide. By Lemma 3.4, we obtain the following global dimension result.

Corollary 4.7. *The space of ω -eigendistributions is one-dimensional and spanned by the distribution*

$$z_0(\omega) = \prod_v z_0(\omega_v).$$

We have just defined a global version of z_0 ; we will now do the same to z .

Definition 4.8.

$$\langle z(\omega), f \rangle = \int_{\mathbb{A}_K^*} f(x) \omega(x) d^*x,$$

provided this integral converges for all f .

By Remark 3.7, (4.3), and appropriate convergence theorems, this converges for ω of exponent greater than 1, and $\langle z(\omega), \bigotimes_v f_v \rangle = \prod_v \langle z(\omega_v), f_v \rangle$. Therefore, for such ω , multiplying the local instances of (3.5) yields

$$(4.9) \quad z(\omega) = z_0(\omega) \cdot \Lambda(\omega),$$

where $\Lambda(\omega) = \prod_v L_v(\omega_v)$.

Note that z and Λ are meromorphic in ω where they are defined, and further that z_0 is holomorphic in ω and nonzero.

4.4. Global Fourier analysis. Let $(\psi_v)_v$ be nontrivial characters of the completions K_v of K , cofinitely many of which have kernel $\mathcal{O}_{K_v}$. By Proposition 2.9, $\prod_v \psi_v$ is a character of $\mathbb{A}_K$, and $y \mapsto \psi(y \cdot -)$ gives an isomorphism of $\mathbb{A}_K$ with its character group; that is, $\mathbb{A}_K$ is self-dual.

If we further require ψ to be trivial on K , this restricts to an isomorphism of K with the character group of $\mathbb{A}_K/K$. It is enough to take on faith for now that such a ψ exists; we will explicitly construct one at the beginning of Section 5, and that this ψ works is [9, Thm 4.1.4] and [9, Lem 4.1.5].

Let ψ be a character of $\mathbb{A}_K$, *trivial on K* . If $f \in S(\mathbb{A}_K)$, define its Fourier transform by $\hat{f}(\xi) = \int_{\mathbb{A}_K} f(x) \psi(x\xi) dx$. By (4.3), the Fourier transform of $\bigotimes_v f_v$ is $\bigotimes_v \hat{f}_v$, defined by the local Fourier transforms given by the local ψ_v s. Therefore the Fourier transform is an automorphism of $S(\mathbb{A}_K)$, and we can normalize dx so $\hat{\hat{f}}(x) = f(-x)$ holds. In fact, this self-dual dx is just the product $\prod_v dx_v$ of the local self-dual measures with respect to the local ψ_v s.

We define the Fourier transform of a distribution λ , again, by $\langle \hat{\lambda}, f \rangle = \langle \lambda, \hat{f} \rangle$. We can now multiply the local instances of Theorem 3.10 and obtain a global analogue.

Theorem 4.10. *There is a nonzero holomorphic factor $\varepsilon(\omega)$ such that $\varepsilon(\omega) \cdot z_0(\omega) = \widehat{z_0(\hat{\omega})}$; $\varepsilon(\omega)$ is given by $\prod_v \varepsilon_v(\omega_v, \psi_v)$. (This is in fact a finite product.)*

Notice $\varepsilon(\omega)$ doesn't depend on ψ by Proposition 3.11.

4.5. Global functional equation. We can try to extend $z(\omega)$ by (4.9), but this requires a meromorphic extension of Λ . Instead, we reverse the situation, extending $z(\omega)$ meromorphically and defining $\Lambda(\omega)$ as the factor making (4.9) true. Once this is done, (4.9) and Theorem 4.10 combine to yield the equation

$$(4.11) \quad z(\omega) = \frac{\Lambda(\omega)}{\Lambda(\hat{\omega}) \cdot \varepsilon(\omega)} \cdot \widehat{z(\hat{\omega})}.$$

We may be tempted to call this the global functional equation. Instead, we will give that name to the following equation, which is the critical computation of Tate's thesis.

Theorem 4.12 (Global functional equation). *z has a meromorphic extension to all quasicharacters satisfying*

$$z(\omega) = \widehat{z(\hat{\omega})}.$$

Corollary 4.13. *Λ has a meromorphic extension satisfying $\Lambda(\omega) = \varepsilon(\omega) \cdot \Lambda(\hat{\omega})$.*

Proof. Combine Theorems 4.11 and 4.12. □

Before proving Theorem 4.12, we need a prerequisite.

Lemma 4.14 (Poisson summation). *For $f \in S(\mathbb{A}_K)$,*

$$\sum_{x \in K} f(x) = \sum_{\xi \in K} \hat{f}(\xi).$$

Proof sketch. Say $\sum_{s \in K} f(x + y)$ converges absolutely and uniformly on compact sets for any $y \in \mathbb{A}_K$, and the same holds for $\hat{f}$. Then this formula follows from repeating the classical proof; see [9, Lem 4.2.4] or [8, Thm 7-7]. Further, any $f \in S(\mathbb{A}_K)$ satisfies these conditions by [8, Lem 7-6]. □

Remark 4.15. The triviality of ψ on K forces the self-dual measure dx to assign the measure 1 to a fundamental domain for K . Poisson summation, in this form, is equivalent to this fact; Tate [9] uses it to prove Poisson summation. Ramakrishnan and Valenza [8] avoid using this fact, instead taking the scaling of dx into account via Fourier inversion.

Theorem 4.16 (Riemann-Roch). *For $f \in S(\mathbb{A}_K)$ and $a \in \mathbb{A}_K^*$,*

$$|a|^{-1} \sum_{x \in K} f(xa^{-1}) = \sum_{\xi \in K} \hat{f}(\xi a).$$

Proof. Apply Lemma 4.14 to $f(xa^{-1})$. □

If K has positive characteristic, the case $f = \bigotimes_v f_v^\circ$ recovers the classical Riemann-Roch theorem for curves over finite fields; see [8, Thm 7.12]. This can therefore be viewed as a restatement and number-theoretic analogue of the classical Riemann-Roch theorem.

Proof of Theorem 4.12. We treat only the case that K is a number field. The positive-characteristic case is similar; see [8, Thm 7-16].

Choose a splitting $\mathbb{A}_K^* = \mathcal{U}_{\mathbb{A}_K} \oplus \text{im } ||$, identifying $\text{im } || = (0, \infty)$ with a subgroup of $\mathbb{A}_K^*$. Let dx be Lebesgue measure, and let $dx_r = dx/x$. There is now a unique measure dx_u on $\mathcal{U}_{\mathbb{A}_K}$ such that d^*x on $\mathbb{A}_K^*$ is the product $dx_u \times dx_r$.

If ω has exponent greater than 1, we see

$$\begin{aligned} \langle z(\omega), f \rangle &= \int_{\mathbb{A}_K^*} f(x) \omega(x) d^*x = \int_0^\infty \int_{\mathcal{U}_{\mathbb{A}_K}} f(xt) \omega(xt) dx_u dt_r \\ &= \int_1^\infty \int_{\mathcal{U}_{\mathbb{A}_K}} f(xt) \omega(xt) dx_u dt_r + \int_0^1 \int_{\mathcal{U}_{\mathbb{A}_K}} f(xt) \omega(xt) dx_u dt_r \\ &= \int_1^\infty \int_{\mathcal{U}_{\mathbb{A}_K}} f(xt) \omega(xt) dx_u dt_r + \int_1^\infty \int_{\mathcal{U}_{\mathbb{A}_K}} f(xt^{-1}) \omega(xt^{-1}) dx_u dt_r. \end{aligned}$$

We treat the second term. Let E be a **fundamental domain** for K^* in $\mathcal{U}_K$; we remind the reader that [9, Sec 4.3] has a construction. Notice

$$\int_{\mathcal{U}_{\mathbb{A}_K}} f(xt^{-1}) \omega(xt^{-1}) dx_u = \int_E \sum_{y \in K^*} f(yt^{-1}) \omega(xt^{-1}) dx_u.$$

We can almost apply Riemann-Roch 4.16; we first need to add the final term $\int_E f(0) \omega(xt^{-1}) dx_u$. Once we have, Riemann-Roch gives

$$\begin{aligned} &\int_{\mathcal{U}_{\mathbb{A}_K}} f(xt^{-1}) \omega(xt^{-1}) dx_u + \int_E f(0) \omega(xt^{-1}) dx_u \\ &= \int_E |x^{-1}t| \sum_{\xi \in K} \hat{f}(\xi x^{-1}t) \omega(xt^{-1}) dx = \int_E \sum_{\xi \in K} \hat{f}(\xi x^{-1}t) \hat{\omega}(x^{-1}t) dx_u. \end{aligned}$$

Reversing steps yields that this is

$$\int_{\mathcal{U}_{\mathbb{A}_K}} \hat{f}(xt) \hat{\omega}(xt) dx_u + \int_E \hat{f}(0) \hat{\omega}(xt) dx_u.$$

We have, therefore,

$$\begin{aligned} \langle z(\omega), f \rangle &= \int_1^\infty \int_{\mathcal{U}_{\mathbb{A}_K}} f(xt) \omega(xt) dx_u dt_r + \int_1^\infty \int_{\mathcal{U}_{\mathbb{A}_K}} \hat{f}(xt) \hat{\omega}(xt) dx_u \\ &\quad - f(0) \int_1^\infty \int_E \omega(xt^{-1}) dx_u dt_r + \hat{f}(0) \int_1^\infty \int_E \hat{\omega}(xt) dx_u dt_r \end{aligned}$$

If ω is unramified, that is if ω is nontrivial on $\mathcal{U}_{\mathbb{A}_K}$, the last two terms are 0. Otherwise, $\omega(x) = |x|^s$ for some $s \in \mathbb{C}$, so these terms are

$$\begin{aligned} &-f(0) \int_1^\infty |t|^{-s} dt_r \int_E dx_u + \hat{f}(0) \int_1^\infty |t|^{1-s} dt_r \int_E dx_u \\ &= - \left(\frac{f(0)}{s} + \frac{\hat{f}(0)}{1-s} \right) \int_E dx_u. \end{aligned}$$

In the notation of [9], we obtain

$$(4.17) \quad \langle z(\omega), f \rangle = \int_1^\infty \int_{\mathcal{U}_{h_K}} f(xt) \omega(xt) dx_u dt_r + \int_1^\infty \int_{\mathcal{U}_{h_K}} \hat{f}(xt) \hat{\omega}(xt) dx_u \\ - \left\{ \left\{ \left(\frac{f(0)}{s} + \frac{\hat{f}(0)}{1-s} \right) \int_E dx_u \right\} \right\},$$

where the expression $\{\{g(s)\}\}$ is 0 if ω is unramified and $g(s)$ if $\omega = ||^s$. (4.17) now converges for *any* ω other than 1 and $||$, not just ω of exponent greater than 1. We can therefore define $z(\omega)$ by (4.17), and the functional equation $z(\omega) = \widehat{z(\hat{\omega})}$ is immediate. $\square$

As a bonus, z has simple poles exactly at 1 and $||$. With the parameterization $||^s$ of the Riemann surface containing these, we can immediately read off the residues as $-\delta_0 \int_E dx_u$ at 1 and $\hat{\delta}_0 \int_E dx_u$ at $||$.

5. COMPUTATIONS

Now we will compute specific examples. We will identify prime integers with the corresponding finite valuations of $\mathbb{Q}$.

If $K = \mathbb{Q}$, we can choose additive characters as follows. Let $\psi_\infty(x) = e^{-2\pi i \cdot x}$. For a finite prime p , $\mathbb{Q}_p/\mathbb{Z}_p$ is (algebraically) isomorphic to $\mathbb{Z}[1/p]/\mathbb{Q}$, which (algebraically) embeds into $\mathbb{R}/\mathbb{Z}$. Composing these gives a homomorphism $\lambda : \mathbb{Q}_p/\mathbb{Z}_p \rightarrow \mathbb{R}/\mathbb{Z}$. Let $\psi_p(x) = e^{2\pi i \cdot \lambda(x)}$; this is an additive character of conductor 0. Every ψ_p is unramified, hence $\psi = \prod_p \psi_p$ gives a character of $\mathbb{A}_{\mathbb{Q}}$ trivial on $\mathbb{Q}$.

For an arbitrary number field K and a valuation v of K lying over a valuation p of $\mathbb{Q}$, we can define ψ_v by composing ψ_p with the trace map $K_v \rightarrow \mathbb{Q}_p$. If p is finite, this has kernel $\text{diff}^{-1}(K_v/\mathbb{Q}_p)$. Cofinitely many of these differents are 1, so $\psi = \prod_v \psi_v$ gives a character of $\mathbb{A}_K$, which is also trivial on K .

For finite v , let ν be the conductor of ψ_v ; that is, $\mathfrak{m}_v^{-\nu} = \text{diff}(\mathcal{O}_{K_v}/\mathbb{Z}_p)$. As we checked in Subsection 3.4, for finite v , the measure on K_v which is self-dual with respect to ψ_v assigns $\mathcal{O}_{K_v}$ the measure $|\pi_v|_v^{-\nu/2}$.

For more details, see [9, Sec 2.2] and [9, Lem 4.1.5]. We will now fix this choice of ψ for computation of ε .

5.1. Warm-up: Riemann zeta function. Let K be $\mathbb{Q}$. For $s \in \mathbb{C}$, let ω_s be $||^s$, so $\widehat{\omega_s} = ||^{1-s}$. The local factors of ω_s are then $||_v^s$. $\Lambda(\omega_s)$ is the Riemann zeta function $\zeta(s)$ times the extra factor $L_\infty(\omega_s) = \pi^{-s/2} \Gamma(\frac{s}{2})$. Our extension of Λ thus gives an extension of ζ . For our functional equation, every ε_p is 1, so we obtain

$$\Lambda(\omega_s) = \Lambda(\omega_{1-s});$$

that is,

$$(5.1) \quad \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \pi^{\frac{s-1}{2}} \Gamma\left(\frac{1-s}{2}\right) \zeta(1-s).$$

This is the classical functional equation for ζ .

5.2. Dirichlet L -functions. Let χ be a primitive Dirichlet character mod m . As we have already described (Construction 2.15), there is an idèlic character ω , trivial on $\mathbb{Q}^*$, such that $L_p(\omega \cdot ||^s) = \frac{1}{1-\chi(p) \cdot p^{-s}}$ for finite primes p . Therefore

$$\Lambda(\omega \cdot ||^s) = L(s, \chi) \cdot L_\infty(\omega \cdot ||^s)$$

for s with $\text{Re}(s) > 1$. This equation gives a meromorphic extension of $L(s, \chi)$, and by Corollary 4.13 it satisfies

$$L(s, \chi)L_\infty(\omega \cdot ||^s) = \varepsilon(\omega \cdot ||^s)L(1-s, \chi^{-1})L_\infty(\omega^{-1} \cdot ||^{1-s}).$$

We will now compute our local ε factors. For finite p relatively prime to m , ω_p is unramified and ψ_p has conductor $\nu = 0$, so by (3.13) $\varepsilon_p(\omega_p \cdot ||_p^s, \psi_p) = 1$.

As ω_∞ is a character of $\mathbb{R}^*$, we only have two choices for ω_∞ , determined by the image of -1 . $\omega_\infty(-1) = \chi(-1)^{-1} = \chi(-1)$, giving $\varepsilon_\infty(\omega_\infty \cdot ||_\infty^s, \psi_\infty) = \begin{cases} 1 & \text{if } \chi(-1) = 1 \\ i & \text{if } \chi(-1) = -1 \end{cases}$.

We now need only to deal with the remaining ε factors. The conductor c_p of ω_p is the multiplicity with which p divides m . Computing, we obtain

$$\begin{aligned} \prod_{p|m} \varepsilon_p(\omega_p \cdot ||_p, \psi_p) &= \prod_{p|m} \int_{p^{-c_p} \mathbb{Z}_p^*} \omega_p^{-1}(x) |x|_p^{-s} \psi_p(x) dx \\ &= m^{-s} \int_{m^{-1} \cdot \prod_{p|m} \mathbb{Z}_p^*} \omega^{-1} \left(\prod_{p|m} \iota_p(x_p) \right) \cdot \prod_{p|m} \psi_p(x_p) dx \\ &= m^{-s} \sum_{x \in (\mathbb{Z}/m)^*} \chi(x) \cdot \prod_{p|m} e^{2\pi i \cdot x_p/m} = m^{-s} \sum_{x \in (\mathbb{Z}/m)^*} \chi(x) \cdot e^{2\pi i \cdot x/m}, \end{aligned}$$

where x_p denotes the image of x under the composition $\mathbb{Z}/m \rightarrow \mathbb{Z}/p^{c_p} \rightarrow \mathbb{Z}/m$ of maps given by the Chinese remainder theorem. We collect our final functional equation: it is

(5.2)

$$L(s, \chi)L_\infty(\omega \cdot ||^s) = \varepsilon_\infty L(1-s, \chi^{-1})L_\infty(\omega^{-1} \cdot ||^{1-s}) m^{-s} \sum_{x \in (\mathbb{Z}/m)^*} \chi(x) \cdot e^{2\pi i \cdot x/m},$$

where

$$\begin{aligned} \varepsilon_\infty &= \begin{cases} 1 & \text{if } \chi(-1) = 1 \\ -i & \text{if } \chi(-1) = -1 \end{cases}, \\ L_\infty(\omega \cdot ||^s) &= \begin{cases} \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) & \text{if } \chi(-1) = 1 \\ \pi^{-\frac{s+1}{2}} \Gamma\left(\frac{s+1}{2}\right) & \text{if } \chi(-1) = -1 \end{cases}, \\ \text{and } L_\infty(\omega^{-1} \cdot ||^{1-s}) &= \begin{cases} \pi^{\frac{s-1}{2}} \Gamma\left(\frac{1-s}{2}\right) & \text{if } \chi(-1) = 1 \\ \pi^{\frac{s-2}{2}} \Gamma\left(\frac{2-s}{2}\right) & \text{if } \chi(-1) = -1 \end{cases}. \end{aligned}$$

5.3. Hecke characters. Let K be a number field. Fix an ideal $\mathfrak{f}$ of $\mathcal{O}_K$, and let $\mathfrak{f} = \prod_{\mathfrak{p}} \mathfrak{p}^{c_{\mathfrak{p}}}$ be its factorization into prime ideals. For $x \in \mathcal{O}_K$, let $\tilde{x}$ denote the residue of x mod $\mathfrak{f}$.

Definition 5.3. A *Hecke character mod $\mathfrak{f}$* is a character χ of the group of fractional ideals relatively prime to $\mathfrak{f}$ satisfying the following condition: a character ω_{∞} of $\prod_{v \text{ arch.}} K_v^*$ exists such that, for all $x \equiv 1 \pmod{\mathfrak{f}}$, $\chi((x)) = \omega_{\infty}^{-1}(x)$. Notice χ uniquely specifies ω_{∞} .

Definition 5.4. A Hecke character mod $\mathfrak{f}$ is called *primitive* if it is not induced (by restriction) by a Hecke character mod $\mathfrak{f}'$ for any $\mathfrak{f}'$ properly dividing $\mathfrak{f}$.

If χ is a Hecke character mod $\mathfrak{f}$, $\chi_{\mathfrak{f}}(\tilde{x}) = \chi((x)) \cdot \omega_{\infty}(x)$ gives a well-defined character of $(\mathcal{O}_K/\mathfrak{f})^*$. Hence we can define Hecke characters as characters which factor as $\chi_{\mathfrak{f}}(\tilde{x}) \cdot \omega_{\infty}^{-1}(x)$, for some $\chi_{\mathfrak{f}}$ and ω_{∞} , on principal ideals coprime to $\mathfrak{f}$.

Given a Hecke character χ mod $\mathfrak{f}$, we can define an *L-function*

$$(5.5) \quad L(s, \chi) = \sum_I \frac{\chi(I)}{\|I\|^s} = \prod_{\mathfrak{p}} \frac{1}{1 - \chi(\mathfrak{p})\|\mathfrak{p}\|^{-s}},$$

where the sum is taken over ideals of $\mathcal{O}_K$ relatively prime to $\mathfrak{f}$, the product is taken over primes of $\mathcal{O}_K$ not dividing $\mathfrak{f}$, and $\|I\| = |\mathcal{O}_K/I|$. As in the Dirichlet case, we want to interpret this as a product of most of the factors of $\Lambda(\omega \cdot \| \cdot \|^s)$ for an appropriate ω . We give a very explicit and computational, yet not very clean, construction of ω .

- If $\mathfrak{p}$ is a prime not dividing $\mathfrak{f}$, the requirement $L_{v_{\mathfrak{p}}}(\omega_{v_{\mathfrak{p}}} \cdot \| \cdot \|_{v_{\mathfrak{p}}}^s) = \frac{1}{1 - \chi(\mathfrak{p})\|\mathfrak{p}\|^{-s}}$ forces $\omega_{v_{\mathfrak{p}}}$ to be unramified with $\omega_{v_{\mathfrak{p}}}(\pi_v) = \chi^{-1}(\mathfrak{p})$.
- For infinite v , let ω_v be the factor of ω_{∞} at v .
- For every remaining valuation v , let ω_v be the factor of ω_{ram} at v , where $\omega_{\text{ram}} : \prod_{\mathfrak{p}|\mathfrak{f}} K_{v_{\mathfrak{p}}}^* \rightarrow S^1$ is defined as follows. First, define ω_{ram} on $\prod_{\mathfrak{p}|\mathfrak{f}} \mathcal{U}_{v_{\mathfrak{p}}}$ as the pullback of $\chi_{\mathfrak{f}}^{-1}$ under the quotient $\prod_{\mathfrak{p}|\mathfrak{f}} \mathcal{U}_{v_{\mathfrak{p}}} \rightarrow (\mathcal{O}_K/\mathfrak{f})^*$. The factorization $\chi((x)) = \chi_{\mathfrak{f}}(\tilde{x}) \cdot \omega_{\infty}^{-1}(x)$ now guarantees that, for whatever extension ω_{ram} we choose, $\omega(x) = \prod_v \omega_v(x_v)$ is trivial on every $\alpha \in K^*$ coprime to $\mathfrak{f}$.

We now need to extend ω_{ram} to the rest of $\prod_{v|\mathfrak{f}} K_v^* \rightarrow S^1$ to make ω trivial on all of K^* . This is obtained by the weak approximation theorem 2.12. (Continuity follows by translating to the case $x \in \prod_{v|\mathfrak{f}} \mathcal{U}_v$, where we have already explicitly constructed a continuous ω_{ram} .)

This gives a map from Hecke characters χ to characters ω of $\mathbb{A}_K^*$.

We can go the other way: let ω be a character of $\mathbb{A}_K^*$ and $\mathfrak{f}$ an ideal such that, for every prime $\mathfrak{p}$, $\mathfrak{p}^{c_{\mathfrak{p}}}|\mathfrak{f}$ (where $c_{\mathfrak{p}}$ is the conductor of ω_v). Then we can define a Hecke character χ by letting $\chi(\mathfrak{p}) = \omega_{v_{\mathfrak{p}}}(\pi_{v_{\mathfrak{p}}})$ and extending by linearity. The associated ω_{∞} is then $\prod_{v \text{ arch.}} \omega_v$.

These are inverse bijections between primitive characters and characters of $\mathbb{A}_K^*$. Hence, the classical and idèlic pictures are equivalent. $\chi_{\mathfrak{f}}^{-1}$ is (with some obvious identifications) the product of the ramified ω_v s, restricted to their $\mathcal{U}_v$ s. ω_{∞} is ω_{∞} . For the unramified ω_v s, $\omega_v(\pi_v)$ encode χ itself. The remaining $\omega_v(\pi_v)$ s, though not seen in the classical picture, are there to make ω trivial on K^* .

As a corollary of the equivalence between the classical and idèlic pictures, for K a number field, the content of Corollary 4.13 is exactly the functional equation

for $L(s, \chi)$ for primitive Hecke characters χ . In particular, if we define $\Lambda(s, \chi) = \Lambda(\omega \cdot ||^s)$, where χ and ω are corresponding Hecke and idèlic characters, we obtain

$$(5.6) \quad \Lambda(s, \chi) = L(s, \chi) \cdot \prod_{v \text{ arch.}} L_v(\omega_v \cdot ||_v^s)$$

for $\text{Re}(s) > 1$. The meromorphic extension and functional equation of Λ yields the same for L .

Finally, we point out that Dirichlet characters are the case $K = \mathbb{Q}$; $\omega_\infty = 1$ corresponds to $\chi(-1) = 1$ and $\omega_\infty = \text{sgn}$ corresponds to $\chi(-1) = -1$. The distinction between these two kinds of Dirichlet characters comes (in this view) from their having different infinite parts.

5.4. Example: Hecke characters of $\mathbb{Q}(\sqrt{5}) \bmod (\sqrt{5})$. To illustrate the situation just described, we will work out the exercise on [5, Page 130]: compute the functional equations of $L(s, \chi)$ for Hecke characters of $\mathbb{Q}(\sqrt{5}) \bmod \mathfrak{f} = (\sqrt{5})$.

First, we will determine which ω_∞ s are allowed. There are two infinite places of $\mathbb{Q}(\sqrt{5})$, namely $|a + b\sqrt{5}|_{\infty_1} = |a + b\sqrt{5}|$ and $|a + b\sqrt{5}|_{\infty_2} = |a - b\sqrt{5}|$, and both of these are real. ω_∞ is a character of $\mathbb{R}^* \times \mathbb{R}^*$; therefore we may write it uniquely as $\omega_\infty(x_1, x_2) = |x_1|^{a_1} |x_2|^{a_2} \text{sgn}(x_1)^{s_1} \text{sgn}(x_2)^{s_2}$ for $a_1, a_2 \in \mathbb{R}$ and $s_1, s_2 \in \mathbb{Z}/2$.

For ω_∞ to correspond to a Hecke character χ , $\omega_\infty^{-1}(x) = \chi((x))$ for $x \equiv 1 \pmod{(\sqrt{5})}$ forces ω_∞ to be trivial on units of $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$ which are congruent to 1 mod $(\sqrt{5})$. As -1 and $\frac{1+\sqrt{5}}{2}$ generate the units of $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$, the group of units congruent to 1 mod $(\sqrt{5})$ is principal and generated by $-\left(\frac{1+\sqrt{5}}{2}\right)^2$. ω_∞ is trivial on $-\left(\frac{1+\sqrt{5}}{2}\right)^2$ if and only if $\left|\frac{1+\sqrt{5}}{2}\right|^{2(a_1-a_2)} \cdot (-1)^{s_1+s_2} = 1$; that is, some $n \in \mathbb{Z}$ exists such that $a_1 - a_2 = \pi i n / 2 \log\left(\frac{1+\sqrt{5}}{2}\right)$ and $s_1 + s_2 \equiv n \pmod{2}$.

Say we have such an ω_∞ . As $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$ is a PID and its units surject to those of $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}/(\sqrt{5}) \cong \mathbb{F}_5$, we can retrieve the χ with this ω_∞ by defining $\chi((x/y)) = \omega_\infty^{-1}(x) \cdot \omega_\infty(y)$ for $x, y \equiv 1 \pmod{(\sqrt{5})}$. This gives a well-defined character on all ideals coprime to $(\sqrt{5})$. Computing the corresponding $\chi_{\mathfrak{f}}$, we see

$$\chi_{\mathfrak{f}}(\tilde{3}) = \chi_{\mathfrak{f}}\left(\widetilde{\frac{1+\sqrt{5}}{2}}\right) = \omega_\infty\left(\frac{1+\sqrt{5}}{2}\right) \chi\left(\left(\frac{1+\sqrt{5}}{2}\right)\right) = \omega_\infty\left(\frac{1+\sqrt{5}}{2}\right),$$

and $\tilde{3}$ generates $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}/(\sqrt{5}) \cong \mathbb{F}_5^*$.

Let ω be the idèlic character corresponding to χ . At every finite place v other than $v_{\sqrt{5}}$, $|\sqrt{5}|_v = 1$ and ω_v is unramified, so $\omega_v(\sqrt{5}) = 1$. As $\omega(\sqrt{5}) = 1$, $\omega_{v_{\sqrt{5}}}(\sqrt{5}) = \omega_\infty^{-1}(\sqrt{5})$. This, that $\omega_{v_{\sqrt{5}}}$ has conductor 1, and $\omega_{v_{\sqrt{5}}}(3) = \chi_{\mathfrak{f}}^{-1}(\tilde{3}) = \omega_\infty^{-1}\left(\frac{1+\sqrt{5}}{2}\right)$ completely determine $\omega_{v_{\sqrt{5}}}$.

The description of ω_v for other finite v is straightforward. Pick a generator x of $\mathfrak{p}_v$ which is congruent to 1 mod $(\sqrt{5})$. ω_v is unramified, and $\omega(x) = 1$ forces $\omega_v(\pi_v) = \omega_v(x) = \omega_\infty^{-1}(x)$ to hold.

We can now finally compute all our local ε factors. $\text{diff}(\mathcal{O}_{\mathbb{Q}(\sqrt{5})}/\mathbb{Z}) = (\sqrt{5})$, so $\varepsilon_v(\omega_v \cdot ||_v^s, \psi_v) = 1$ for every finite v other than $v_{\sqrt{5}}$. From (3.14), we obtain

$$\varepsilon_{v_{\sqrt{5}}}(\omega_{v_{\sqrt{5}}} \cdot ||_{v_{\sqrt{5}}}^s, \psi_{v_{\sqrt{5}}}) = 5^{1/2-2s} \omega_{\infty}^{-1}(5) \sum_{x \in \mathbb{F}_5^*} \chi_{\mathfrak{f}}(x) e^{4\pi i \cdot x/5}.$$

Our infinite ε -factors are

$$\varepsilon_{\infty_i}(\omega_{\infty_i} \cdot ||_{\infty_i}^s, \psi_{\infty_i}) = \begin{cases} 1 & \text{if } a_i = 0 \\ i & \text{if } a_i = 1 \end{cases}.$$

Our functional equation is therefore

$$\Lambda(s, \chi) = \Lambda(1-s, \chi^{-1}) \varepsilon_{\infty} \cdot 5^{1/2-2s} \omega_{\infty}^{-1}(5) \sum_{x \in \mathbb{F}_5^*} \chi_{\mathfrak{f}}(x) e^{4\pi i \cdot x/5},$$

where

$$\begin{aligned} \Lambda(s, \chi) &= L(s, \chi) \cdot L_{\infty_1}(\omega_{\infty_1} \cdot ||_{\infty_1}^s) \cdot L_{\infty_2}(\omega_{\infty_2} \cdot ||_{\infty_2}^s), \\ \Lambda(1-s, \chi^{-1}) &= L(1-s, \chi^{-1}) \cdot L_{\infty_1}(\omega_{\infty_1}^{-1} \cdot ||_{\infty_1}^{1-s}) \cdot L_{\infty_2}(\omega_{\infty_2}^{-1} \cdot ||_{\infty_2}^{1-s}), \\ \varepsilon_{\infty} &= \begin{cases} 1 & \text{if } a_1 = a_2 = 0 \\ -i & \text{if } a_1 \neq a_2 \\ -1 & \text{if } a_1 = a_2 = 1 \end{cases}, \end{aligned}$$

and L_{∞_1} and L_{∞_2} are as in 5.2.

5.5. Final remarks. We refer the reader to [8] for classical applications; among these are the Tchebotarev density theorem.

If K is a field of positive characteristic containing $\mathbb{F}_q$, then $\zeta_K(s) = \Lambda(||^s)$ is in fact a *rational* function in q^{-s} ! See Exercise 22 in Chapter 7 of [8] for a proof. The properties of this rational function are generalized by Weil's celebrated conjectures, now resolved.

REFERENCES

- [1] Bump, Daniel A. *Automorphic Forms and Representations*. Cambridge: Cambridge University Press, 1998.
- [2] Cassels, J. W. S. "Global Fields." In *Algebraic Number Theory*, edited by J. W. S. Cassels and A. Fröhlich, 42-84. London: Academic Press Inc., 1967.
- [3] Folland, Gerald B. *Real Analysis: Modern Techniques and Their Applications*. New York: John Wiley & Sons, Inc., 1999.
- [4] Hecke, E. "Eine neue Art von Zetafunktionen und ihre Beziehungen zur Verteilung der Primzahlen." *Mathematische Zeitschrift*, Issue 1-2, Volume 6 (March 1920): 11-51.
- [5] Kudla, Stephen S. "Tate's Thesis." In *An Introduction to the Langlands Program*, edited by Joseph Bernstein and Stephen Gelbart, 109-131. New York: Springer Science+Business Media, 2004.
- [6] Lang, Serge. *Algebraic Number Theory*. New York: Springer Science+Business Media, 1986.
- [7] Neukirch, Jürgen. "Zeta Functions and L -series." In *Algebraic Number Theory*, 419-549. Translated by Norbert Schappacher. New York: Springer Science+Business Media, 1999.
- [8] Ramakrishnan, Dinakar, and Valenza, Robert J. *Fourier Analysis on Number Fields*. London: Academic Press Inc., 1967.
- [9] Tate, J. T. "Fourier Analysis in Number Fields and Hecke's Zeta-Functions." In *Algebraic Number Theory*, edited by J. W. S. Cassels and A. Fröhlich, 305-347. London: Academic Press Inc., 1967.
- [10] Weil, André. "Fonction zêta et distributions." *Séminaire Bourbaki*, Volume 9 (1964-1966), Talk no. 312, 523-531.